

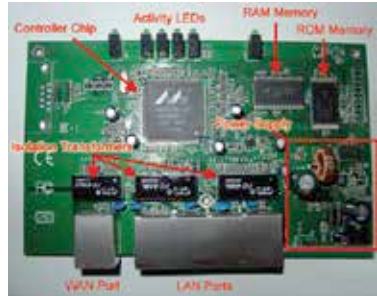
operate with a Timer and a Temperature sensor (to measure the warmth of the water in the tub). On the other hand, a security system or car navigation control system might need many more.

Your phone has a few other (and different) kinds of sensors such as a Proximity sensor, Accelerometer, GPS sensor, Bluetooth, Wi-Fi and so on. All together, they feed data to

the main system (the controller system to which all other components are connected and governed by) which then, based on your current tasks, would take actions. For example, if you're talking on your smartphone, then the input from the proximity sensor will be used to switch off the display while at other times (when you're not talking over the phone), it will remain irrelevant. Similarly, Light sensors are used to dim or brighten your display, while a Wi-Fi adapter can show you the signal quality and available networks.

There are even more types of sensors which can be connected to/used in an embedded system but that would depend on the requirements. For example, a proximity sensor is useless in a microwave oven control system. The requirement of the work environment where the device is to be deployed determines the inclusion of sensors on an embedded system.

However, more the number of sensors more complicated the software and more equipped the hardware must be. This makes you aware of a hardware design constraint of not having enough communication ports through which other devices could communicate. However in the end, the specific design constraints depend on the purpose of the device. Since, we can't talk about all the systems in this tiny booklet; let's go on a walk-through of the common architectural problems and constraints.



Embedded System RAM too is attached to the main board

Architectural considerations

There are a number of requirements which generally pose a challenge to all types of embedded devices.

Component composition

The system architecture should be able to support the composition of a large system using smaller components and dedicated sub-systems. This